Kanghwi Lee

Audio & Speech ML | Generative & Representation Learning
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Profile

Audio & speech ML researcher-engineer and Ph.D. candidate at ETH Zurich with 4+ years of experience building machine-learning workflows for real-world acoustic recordings, developing methods that transfer across human and animal audio. Incoming NAVER Cloud research intern working on TTS audio watermarking for synthetic-speech provenance and safety (Summer–Fall 2026). Work spans foundation-model adaptation, scalable audio curation, and label-free generative modeling over learned audio representations, with a strong Python/PyTorch background across high-throughput acoustic data processing, representation learning, efficient inference, and quantitative audio-model evaluation. Graduating early 2027; open to research-scientist, research-engineer, and applied-scientist roles in audio and speech ML; also interested in multimodal directions building on this work.

Relevant Impact Snapshot

Generative audio representations	Introduced Trajectory Variance, a label-free framework for estimating how individual audio events would change across developmental time using counterfactual generative modeling over learned representations; accepted to Interspeech 2026.
Scalable audio curation	Built ML-assisted audio curation workflows for 80,000+ hours of continuous recordings, shifting human effort from item-by-item labeling toward cluster-level review, acoustic-diversity inspection, and outlier correction; benchmarked on 1,454 vocalizations with approximately 50% annotation-time reduction and ARI = 0.73 / NMI = 0.81.
Foundation-model adaptation	Contributed to WhisperSeg (ICASSP 2024), a Whisper-based system for automatic voice activity detection in long, noisy human and animal recordings, reaching segment-wise F1 up to 0.9966.

Selected Audio ML and Engineering Work

Trajectory Variance – Label-Free Framework for Audio Change Over Time — Accepted to Interspeech 2026 | code
PyTorch, VAEs, Flow Matching (explored), Learned Audio Representations, Optimal Transport, Counterfactual Generation

- Conceived, implemented, and evaluated Trajectory Variance, a label-free framework for estimating how individual audio events would change across developmental time, using approximately 680K vocalizations.
- Designed the counterfactual pipeline with VAE latents, age-conditioned displacement modeling, OT-based pairing, and controlled baseline comparisons.

Large-Scale Audio Curation with Human-in-the-Loop Review — Applied Sciences 2026 | paper | code
Whisper Embeddings, Clustering, Human-in-the-Loop Review, Large-Scale Audio Data

- Designed and evaluated a semi-automated workflow using Whisper encoder embeddings, clustering, and interactive review to organize real-world vocalization data.
- Built reusable curation logic for cluster-structure review, acoustic-diversity inspection, and outlier correction in downstream large-scale audio analysis.

WhisperSeg – Foundation-Model Adaptation for Long-Context Voice Activity Detection — ICASSP 2024 | paper
Whisper, Transformers, Voice Activity Detection, Noisy Recordings, Efficient Inference

- Contributed to the published WhisperSeg system for long-context voice activity detection and used its segmentation outputs in downstream high-throughput audio workflows.
- Worked on dataset processing, baseline evaluation, and cross-domain analysis showing how Whisper-based representations transfer beyond standard ASR to long, noisy human and animal recordings.

Technical Skills

Machine Learning: PyTorch, Transformers, Whisper, Foundation-Model Adaptation, VAEs, Representation Learning, Generative Modeling (incl. Flow-Based), Counterfactual Generation, Optimal Transport
Audio Processing: Voice Activity Detection (VAD), Audio Event Segmentation, Spectrogram Analysis, Acoustic Feature Extraction
Data & Evaluation: Large-Scale Audio Curation (ML-Assisted, Human-in-the-Loop), Clustering, Quantitative Audio-Model Evaluation
Software & Infrastructure: Python, NumPy, Pandas, Hugging Face, Git, Linux, HPC, Multi-GPU Training, CTranslate2

Selected Publications

- K. Lee, "Trajectory Variance: An Unsupervised Measure of Developmental Vocal Plasticity," accepted to Interspeech 2026.
- K. Lee et al., "Benchmarking Automated and Semi-Automated Vocal Clustering Methods," Applied Sciences, 2026.
- N. Gu, K. Lee, et al., "Positive Transfer of the Whisper Speech Transformer to Human and Animal Voice Activity Detection," ICASSP 2024.

Education

ETH Zurich – Ph.D. Candidate, Institute of Neuroinformatics — 2022–Present
Audio Machine Learning and Computational Neuroscience
University of Zurich – M.Sc., Neural Systems and Computation, GPA 5.5/6.0 — 2020–2022
Korea University – B.S. Computer Science; B.Econ. Statistics, GPA 4.02/4.50 — 2014–2020